

Grade 7
Unit 5
Lesson 7 Practice

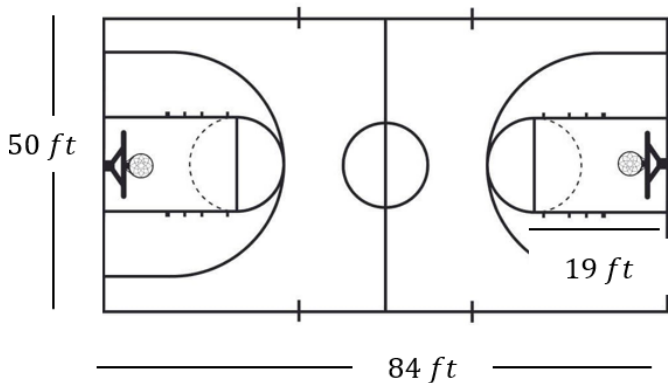
Name Answer Key
Date _____
Period _____

1. The regulation size for a high school basketball court is 84 feet long by 50 feet wide. The foul line is 19 feet from one side of the court. Johnny wants to create a scale drawing of his middle school's regulation basketball court. What steps should Johnny take to draw his scale model?

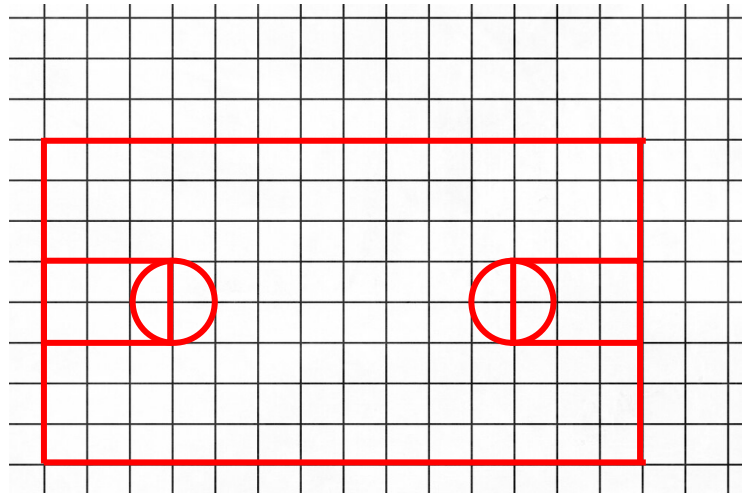
Answers will vary.

Choose a scale factor, less than 1 so your model is smaller than the actual. Then, multiply the scale factor by all dimensions

2. Using a scale factor of $\frac{1}{6}$ create a scale drawing of Johnny's high school regulation basketball court.



Scale drawing of the: _____



3. For your scale drawing in #2, one gridline = 6 ft
4. How would your drawing in #2 change if a scale factor of $\frac{4}{5}$ were used instead of $\frac{1}{6}$?
Each gridline would be 1.25 ft.
5. When creating a scale drawing of a court, property, or statue, why is it important to use a scale factor that is a fraction less than 1?
Answers will vary.
A scale factor of 1 would represent the figure as it is now. So, to create a scale drawing, you would want it to be smaller than the original so the scale factor should be less than 1.

6. A scale on a treasure map shows that 3 centimeters represent 8 meters. The treasure is 200 meters from the starting point. How many centimeters (on the map) are represented by the distance from the starting point to the treasure?

$$\frac{3}{8} \times \frac{25}{25} = \frac{75}{200}$$

75 cm

7. A scale on a map of Florida shows that 1 inches represent 5 miles. Johnny measures the distance from Tallahassee to Orlando (on the map) as 51.2 inches. Based on this map, how many miles is Tallahassee from Orlando?

$$\frac{1 \text{ in}}{5 \text{ mi}} \times \frac{51.2}{51.2} = \frac{51.2 \text{ in}}{256 \text{ miles}}$$

256 miles

Triangle A is shown below. The slanted side of the triangle equals 10 cm. Each gridline represents 2 centimeters (cm). Use this figure to answer questions 8-10.

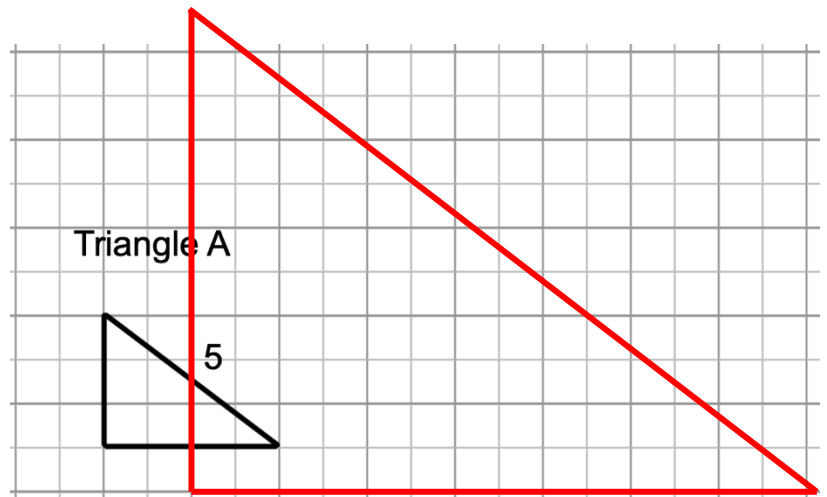
8. Draw a scaled copy of Triangle A that has a perimeter of 84 cm.

$$6 + 8 + 10 = 24$$

$$84 \div 24 = 3.5$$

$$6 \cdot 3.5 = 21; 8 \cdot 3.5 = 28;$$

$$10 \cdot 3.5 = 35$$



9. What is the scale factor of the drawing in #8.

$$\frac{84}{24} = 3.5$$

10. How would the drawing and scale factor change if the perimeter of the copy was decreased to 25.2 cm?

Answers may vary.

The scale factor would decrease as well, $\frac{25.2}{24} = 1.05$, and the scaled copy wouldn't be much larger than the original.